

Methods Practice SAC1 - Part A

By Jesse M

No notes, No CAS

5 mins reading, 50 mins writing,

32 marks

Question 1 (3 marks)

VCAA 2018 Exam 1 Q1

- a. If $y = (-3x^3 + x^2 - 64)^3$, find $\frac{dy}{dx}$.

1 mark

$$\begin{aligned}\text{Let } u &= -3x^3 + x^2 - 64 \Rightarrow y = u^3 \\ \Rightarrow \frac{du}{dx} &= -9x^2 + 2x \quad \Rightarrow \frac{dy}{du} = 3u^2\end{aligned}$$

$$\begin{aligned}\Rightarrow \frac{dy}{dx} &= \frac{du}{dx} \frac{dy}{du} = (-9x^2 + 2x)(3u^2) \\ &= 3(-9x^2 + 2x)(-3x^3 + x^2 - 64)^2 \quad \text{1A}\end{aligned}$$

- b. Let $f(x) = \frac{e^x}{\cos(x)}$.
Evaluate $f'(\pi)$.

2 marks

$$\begin{aligned}f'(x) &= \frac{(e^x)(\cos x) - (e^x)(-\sin x)}{(\cos x)^2} \quad \text{1M} \\ &= \frac{e^x(\cos x + \sin x)}{\cos^2 x} \\ \Rightarrow f'(\pi) &= \frac{e^\pi(\cos \pi + \sin \pi)}{\cos^2 \pi} = \frac{e^\pi(-1 + 0)}{(-1)^2} \\ &= -e^\pi \quad \text{1A}\end{aligned}$$

Question 2 (3 marks)

State the transformation that takes the graph of:

$$y = 2\sqrt{2-x} + 3 \quad \text{to} \quad y = -5\sqrt{x} + 6$$

Give your answer in the form:

$$(x, y) \rightarrow (ax + b, cy + d)$$

original: $y = 2\sqrt{2-x} + 3$

Image: $y' = -5\sqrt{x'} + 6$

$$\Leftrightarrow \frac{y-3}{2} = \sqrt{2-x} \quad \leftarrow \text{IM} \rightarrow$$

$$\Leftrightarrow \frac{y'-6}{-5} = \sqrt{x'}$$

$$\rightarrow 2-x = x' \quad \leftarrow \text{IM} \rightarrow \rightarrow \frac{y-3}{2} = \frac{y'-6}{-5}$$

$$\Rightarrow x' = -x + 2$$

$$\Leftrightarrow -5y + 15 = 2y' - 12$$

$$\Leftrightarrow 2y' = -5y + 27$$

$$\therefore (x, y) \rightarrow (-x+2, -\frac{5}{2}y + \frac{27}{2})$$

$$\Rightarrow y' = -\frac{5}{2}y + \frac{27}{2}$$

Question 3 (5 marks)

VCAA 2018 Exam 1 Q7

Let P be a point on the straight line $y = 2x - 4$ such that the length of OP , the line segment from the origin O to P , is a minimum.

- a. Find the coordinates of P .

3 marks

Let $P = (k, 2k-4)$, and $d(k)$ be the distance from O to P .

$$\begin{aligned} \Rightarrow d(k) &= \sqrt{(k-0)^2 + (2k-4-0)^2} \\ &= \sqrt{k^2 + 4k^2 - 16k + 16} \\ &= \sqrt{5k^2 - 16k + 16}, \text{ IM} \end{aligned}$$

Min at TP of quadratic $\uparrow$, $k = -\frac{(-16)}{2(5)} = \frac{8}{5}$ IM

$$\Rightarrow P = \left(\frac{8}{5}, -\frac{4}{5}\right) \text{ IA}$$

- b. Find the distance OP . Express your answer in the form $\frac{a\sqrt{b}}{b}$, where a and b are positive integers.

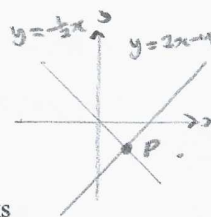
2 marks

Distance O to P is: (use $d(k) = \sqrt{k^2 + (2k-4)^2}$)

$$d\left(\frac{8}{5}\right) = \sqrt{\left(\frac{8}{5}\right)^2 + \left(-\frac{4}{5}\right)^2} \text{ IM}$$

$$= \sqrt{\frac{64}{25} + \frac{16}{25}} = \sqrt{\frac{80}{25}}$$

$$= \sqrt{\frac{16}{5}} = \frac{4}{\sqrt{5}} \quad \text{OR} \quad \frac{4\sqrt{5}}{5} \text{ IA}$$



Question 4 (7 marks)

Neap 2020 Exam 1 Q5

a. Solve $1 - 2\sin(2x) = 0$ for $x \in \left(-\frac{\pi}{2}, \pi\right]$.

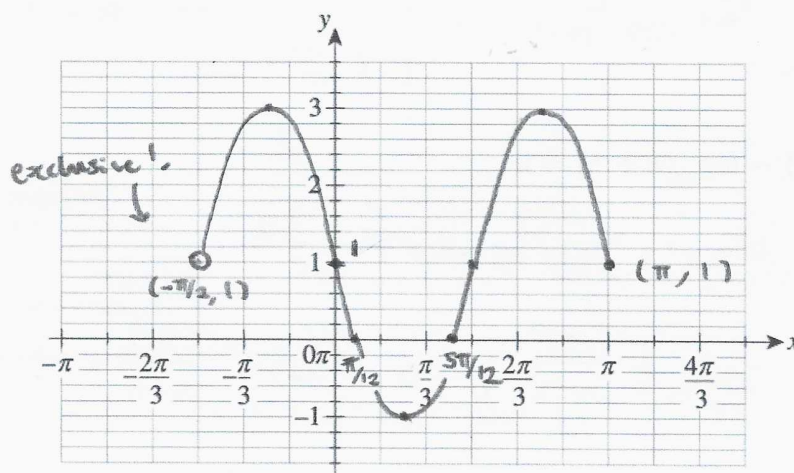
2 marks

$$\begin{aligned} \Leftrightarrow \sin(2x) &= \frac{1}{2}, \quad x \in \left(-\frac{\pi}{2}, \pi\right] \\ \Rightarrow 2x &\in (-\pi, 2\pi] \\ \Rightarrow 2x &= \frac{\pi}{6}, \frac{5\pi}{6} \\ \Rightarrow x &= \frac{\pi}{12}, \frac{5\pi}{12} \quad \text{IA} \end{aligned}$$

Let $f: \left(-\frac{\pi}{2}, \pi\right] \rightarrow \mathbb{R}, f(x) = 1 - 2\sin(2x)$.

- b. Sketch the graph of $y = f(x)$ on the axes below. Label the axes intercepts and endpoints with their coordinates as appropriate.

3 marks



IM: Endpoints
IM: Axes Intercepts
IM: Shape.

- c. Find the equation of the tangent to the graph of $y = f(x)$ at the point where the graph crosses the y-axis.

2 marks

$$\begin{aligned} y - y_1 &= m(x - x_1) \\ \rightarrow f'(x) &= -4\cos(2x) \Rightarrow f'(0) = -4 \\ \Rightarrow \text{Tangent: } y - 1 &= -4(x - 0) \\ \Rightarrow y &= -4x + 1 \quad \text{IA} \end{aligned}$$

Question 5 (5 marks)Solve the following equations for all **real** solutions, $x \in \mathbb{R}$.

a. $3^x + 2 = 15 \times 3^{-x}$

(2 marks)

Let $u = 3^x$.

$$\Rightarrow u + 2 = 15u^{-1} \quad (\times u \text{ on both sides})$$

$$\Leftrightarrow u^2 + 2u = 15 \quad \Leftrightarrow u^2 + 2u - 15 = 0$$

$$\Leftrightarrow (u+5)(u-3) = 0$$

$$\Rightarrow u = -5$$

$$\Rightarrow u = 3$$

$$\Leftrightarrow 3^x = -5 \quad \text{IA}$$

$$\Leftrightarrow 3^x = 3$$

$$\Rightarrow \text{No real sol.}$$

$$\Rightarrow x = 1 \quad \text{IA}$$

b. $2\log_{10}(x) - \log_{10}(x+6) = 0$

(3 marks)

$$\Leftrightarrow \log_{10}(x^2) - \log_{10}(x+6) = 0$$

$$\Leftrightarrow \log_{10}\left(\frac{x^2}{x+6}\right) = 0 \quad \Leftrightarrow \frac{x^2}{x+6} = 1$$

$$\Leftrightarrow x^2 = x+6 \quad \Leftrightarrow x^2 - x - 6 = 0 \quad \text{IM}$$

$$\Leftrightarrow (x-3)(x+2) = 0 \quad \Rightarrow x = 3, \cancel{-2}$$

$$x = 3$$

 $\leftarrow \text{IM} \rightarrow$

$$x = -2$$

$$\Rightarrow \text{LHS} = 2\log_{10}(3) - \log_{10}(9) \\ \rightarrow \text{all good}.$$

$$\Rightarrow \text{LHS} = 2\log_{10}(-2) - \log_{10}(4) \\ \rightarrow \text{negative log!}$$

$$\therefore \underline{x = 3 \text{ only!}} \quad \text{IA}$$

Question 6 (4 marks)Consider the piecewise function, f :

$$f(x) = \begin{cases} x^3 - 3 \cdot x + 9, & x \leq 0 \rightarrow f_1(x) \\ a(x+a), & x > 0 \rightarrow f_2(x) \end{cases}$$

For what value(s) of a is the function f ...a. **Continuous** for all $x \in \mathbb{R}$?

(2 marks)

'Continuous' requires:

$$\rightarrow f_1(0) = f_2(0) \quad 1M$$

$$\Rightarrow 0^3 - 3(0) + 9 = a(0+a)$$

$$\Rightarrow 9 = a^2 \Rightarrow a = \pm 3 \quad 1A$$

b. **Differentiable** for all $x \in \mathbb{R}$?

(2 marks)

'Differentiable' requires continuity, and:

$$\rightarrow f_1'(0) = f_2'(0) \quad 0.5M$$

$$\rightarrow f_1'(x) = 3x^2 - 3, \quad f_2'(x) = \frac{d}{dx}(ax + a^2) = a$$

$$\rightarrow f_1'(0) = f_2'(0)$$

$$\Rightarrow 3(0)^2 - 3 = a \Rightarrow a = -3 \quad 1A$$

Since we know $a = -3$ gives continuity (from a.),

we know it also gives differentiability.

0.5M

} important
to note
 $a = -3$

also gives continuity

Question 7 (5 marks)

Neap 2020 Exam 1 Q7

Let $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = e^{3x} - 2$.

- a. Find the rule and domain of the inverse function f^{-1} .

2 marks

→ Swap x and y :

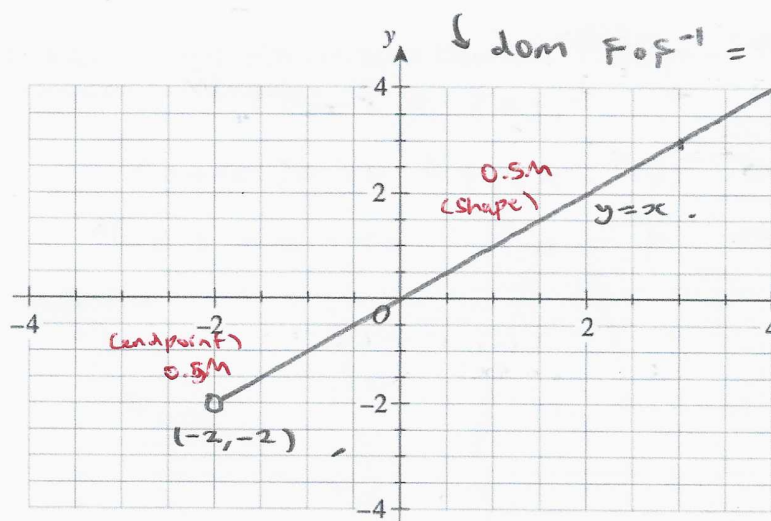
$$\rightarrow x = e^{3y} - 2 \quad \Leftrightarrow \quad x + 2 = e^{3y}$$

$$\Leftrightarrow y = \frac{1}{3} \ln(x+2) \Rightarrow f^{-1}(x) = \frac{1}{3} \ln(x+2) \quad 1A$$

$$\rightarrow \text{dom } f^{-1} = \text{ran } f = (-2, \infty) \quad 1A$$

- b. On the axes below, sketch the graph of $y = f(f^{-1}(x))$ for its maximal domain.

1 mark



dom $f \circ f^{-1} = \text{dom } f^{-1} !!$

(in this case).

→ Also,

$$f(f^{-1}(x)) = x !$$

So we have:

$$y = x, \quad x \in (-2, \infty)$$

- c. Find $f(f^{-1}(3x))$ in the form $\frac{ax+b}{cx+d}$, where $a, b, c, d \in \mathbb{Z}$.

2 marks

$$f(f^{-1}(3x)) = f\left(-\frac{1}{3} \ln(3x+2)\right)$$

$$= e^{3\left(-\frac{1}{3} \ln(3x+2)\right)} - 2 \quad 1M$$

$$= e^{-\ln(3x+2)} - 2$$

$$= (e^{\ln(3x+2)})^{-1} - 2 = \frac{1}{3x+2} - 2$$

$$= \frac{1 - 2(3x+2)}{3x+2} = \frac{-6x-3}{3x+2} \quad 1A$$